Soil Nutrient Depth Profiles

Bayesian Hierarchical Modeling of Carbon, Nitrogen and Phosphorus in Tanzanian Fields

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40	220	3	$\rho = 0.52$	0.39
Fields	Soil Samples	Nutrients (C/N/P)	Carbon at Depth	C/N Coupling

Overview

We study how three key soil nutrients, organic carbon, total nitrogen, and total phosphorus, are distributed with depth across 40 agricultural fields in Tanzania. The question is twofold: how each nutrient changes as we move deeper into the soil, and how this depth profile varies from one field to another. Depth profiles carry agronomic meaning, since whether fertility concentrates near the surface or reaches deeper shapes both crop access and the soil's capacity to store carbon. We fit a **Bayesian hierarchical model** with field level random effects that treats the three nutrients jointly, so their correlations are estimated from the data rather than assumed.

Data

Source: field survey of 40 Tanzanian agricultural fields (Statistics: Case Studies, University of Padova), with 220 soil samples collected at several depths.

Responses: soil organic carbon, total nitrogen, and total phosphorus, all log transformed to stabilize variance and allow an elasticity style reading.

Texture: clay, silt, and sand percentages form a composition, mapped to two isometric log ratio (ILR) contrasts, the natural way to use compositional predictors without spurious constraints.

Covariates: texture (ILR), bulk density, and pH vary within a field with depth; management practices (on farm status, irrigation, fertilisation, natural nitrogen) sit at the field level.

Models

A sequence of Bayesian hierarchical models of increasing richness, all estimated in **Stan** via `cmdstanr`:

- **Random intercept baseline:** each field gets its own average nutrient level.
- **Random depth slopes:** each field also gets its own profile steepness, with the three nutrients tied together through a multivariate LKJ covariance, so intercepts and slopes can correlate across nutrients.
- **Spatial variants:** a Gaussian process on field coordinates tests whether residual spatial structure carries extra information.

Spatial confounding. Because the field level covariates vary smoothly across the landscape, their effect cannot be separated from where each field sits: the covariates and the spatial position are confounded. Adding the Gaussian process term does not improve predictive fit, so the final model keeps the non spatial multivariate specification. Inference uses four parallel chains; all parameters reach $\hat{R} \approx 1.00$ with high effective sample sizes.

Results

Model choice: leave one out cross validation (LOO) favors the multivariate model without the spatial term; the Gaussian process variants do not improve predictive performance.

Carbon is preserved at depth. Within a field, a higher carbon level goes with a flatter carbon profile: where carbon is abundant, it persists deeper into the soil. The profile intercept and slope correlate positively ($\rho = 0.52$, 90% CI $[0.12,\ 0.80]$). The same pattern is absent for nitrogen and only moderate, with wide posterior uncertainty, for phosphorus.

Nutrients move together. At the field level, carbon and nitrogen intercepts are positively correlated ($\rho = 0.39$, 90% CI $[0.04,\ 0.67]$). This coupling is exactly what the joint model is built to capture and would be invisible to three separate single response models.

Texture drives the vertical gradient. Predictive variable selection with `projpred` identifies the fine fraction versus sand ILR contrast as the dominant within field predictor of nutrient decline with depth (carbon effect -0.50, 90% CI $[-0.64,\ -0.36]$; nitrogen -0.28). Bulk density contributes; pH does not.

Key Findings

- **Carbon at depth is special:** richer fields keep flatter carbon profiles, a behaviour unique to carbon among the three nutrients.
- **Joint modeling pays off:** the field level correlation between carbon and nitrogen would be lost by modeling each nutrient on its own.
- **Soil structure leads:** the fine texture contrast is the strongest within field driver, bulk density is secondary, and pH is negligible.
- **Spatial confounding limits causal claims:** field level effects cannot be disentangled from geographic position, so they stay associational, and an explicit spatial term adds no predictive value.

Technologies

Languages & tools: R, Stan (`cmdstanr`), `posterior`, `loo`, `projpred`, `compositions`, `lme4`, `ggplot2`.
Data: field survey of 40 Tanzanian agricultural fields (Statistics: Case Studies, University of Padova).

Future work: collecting more observations would support more complex interactions, also within the random effects themselves; study designs that resolve the spatial confounding; additional nutrients and micronutrients; and out of sample validation on new fields.